Lesson 1 worksheet: My First Traffic Light

Name: _____ Date: _____ **Grown-up who checked my circuit:** _____

> **Keep the USB cable unplugged while you wire.** Plug it in only after a grown-up has checked the circuit. Your teacher puts `main.py` (from this lesson's folder) on the ESP32 before class. No screen or internet needed.

1. Predict (before you build)

The program turns on **red for 5 s**, then **green for 5 s**, then **yellow for 2 s**, and repeats forever.

Question	My guess
Which light comes on **first**?	
How many seconds is one full cycle (red + green + yellow)?	
How many full cycles happen in one minute?	
If one LED is put in **backwards**, what will that light do?	

2. Wire it (tick each box when it is in)

Like water in pipes: the **pin** is a pump, **wires** are pipes, the **resistor** is a narrow pipe that slows the flow, the **LED** is a water wheel that glows and only turns one way.

Light	ESP32 pin → hole	330 Ω resistor (crosses the middle gap)	LED long leg (+)	LED short leg (−)	Black wire to − rail	✔
Red	D25 → b7	d7 to g7	i7	i8	j8 → − rail	☐
Yellow	D26 → b15	d15 to g15	i15	i16	j16 → − rail	☐
Green	D27 → b23	d23 to g23	i23	i24	j24 → − rail	☐
Ground	GND pin (next to D13) → blue − rail					☐

Check: the long leg and the short leg are in **different** rows. Every LED has **its own** resistor.

3. Observe (plug in the USB cable and watch)

Count the seconds out loud ("one elephant, two elephants…") for each light. Do it twice.

	Red	Green	Yellow	Whole cycle
Try 1 (seconds)				
Try 2 (seconds)				

Order I saw (write R, G or Y): ___ → ___ → ___ → ___ → ___ → ___ Was my prediction right? ☐ yes ☐ no

Ask your grown-up to unplug **only** the GND wire for a moment. What happened?

_____

4. Explain (use your own words)

1. Why does each LED need a resistor? _____
2. Why is the short leg in a **different row** from the long leg? _____
3. Why did the lights stop when the GND wire came out? _____
4. The program says `red = Pin(25, Pin.OUT)`. What does the `25` mean? _____

Make it yours: In the lesson you can change the green time (2 to 10 s). If green were **8 s**, how long would one full cycle be? _____ s